Mix It

Deciding what precipitates before you pour, and writing the one equation that says what actually happened.

Name ____________________ Date ____________________ Period ____________________

Demonstration: andresdbp.github.io/resources/demos/mix-it/

1 Before you open anything

Answer from what you already know. Nothing here is graded on being right — the point is to have a commitment on paper that the demonstration can settle later.

PREDICTION

A solution of sodium carbonate, $Na_2CO_3(aq)$, is poured into a solution of potassium chloride, $KCl(aq)$. Carbonates are usually insoluble. Does a solid form? If it does, write its formula; if it does not, write "none."

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Write one or two sentences saying which rule you used and why it applies here.

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2 One pour, start to finish

Open the demonstration. Panel 1 chooses the two bottles, panel 2 takes your prediction and pours, panel 3 builds the three equations underneath.

1. In **1 · Pick two solutions**, choose **sodium chloride + silver nitrate** from the menu.
2. Before touching anything else, read the ion legend under the beaker and record the four ions and how many of each there are. Record **Ions still dissolved** now, before you pour.
3. In **2 · Predict, then pour**, choose one of the three buttons. Write down which one you chose before you look at the feedback.
4. Press **Pour them together** and wait until the beaker stops changing.
5. Fill in the rest of the table from the **Score & count** and **How insoluble is "insoluble"?** panels.

Readout	Value
The four ions, with counts	
Ions still dissolved (before pouring)	
Ions still dissolved (after pouring)	
Ions in the solid	
Solid formed	
K_{sp} at 25 °C	
Molar solubility	

a. Copy the three equations exactly as the demonstration writes them, keeping every state label and every struck-out species.

b. Two ions are struck out in the complete ionic line. Name them, and say what has to be true of an ion for it to be struck out.

c. Add your two count readouts after pouring: *Ions still dissolved* + *Ions in the solid*. Compare the total with the number of ions dissolved before you poured. What does that equality tell you about the spectator ions?

> The beaker draws about two dozen ions. A real 100 mL beaker holds roughly 10^{22} of them — the count on screen is a cartoon, the pairing rules are not.

3 Change one partner at a time

These four pairs come in two matched sets. In each set, one ion is held fixed and only its partner changes. Predict, pour, and record each one.

1. Run **sodium carbonate + calcium chloride**, then **sodium carbonate + potassium chloride**. Both contain CO_3^{2-}.
2. Run **sodium hydroxide + magnesium nitrate**, then **magnesium nitrate + potassium iodide**. Both contain Mg^{2+}.
3. For each, record the solid (or "none"), the net ionic equation, and the two counts after pouring.

Pair	Solid formed	Net ionic equation	Ions still dissolved	Ions in the solid
$Na_2CO_3 + CaCl_2$				
$Na_2CO_3 + KCl$				
$NaOH + Mg(NO_3)_2$				
$Mg(NO_3)_2 + KI$				

a. Rows 1 and 2 both contain carbonate, and carbonate is listed as insoluble. Only one of them precipitated. Using the *Solubility rules used here* table, explain in one or two sentences what decided it.

b. State the rule in your own words: what does a precipitate depend on — the anion alone, the cation alone, or something else? Finish the sentence.

Whether a solid forms is a property of ________________________, not of either ion by itself, because

c. Go back to your prediction in part 1. Row 2 of this table is that exact mixture. Was your prediction right? If it was not, name the specific step in your reasoning that did not hold.

d. In rows 2 and 4 no solid formed. Write what the net ionic equation is in that case, and say why.

4 Working backwards from the net ionic line

A test question rarely hands you the bottles and asks for the equation. It is just as likely to hand you the equation and ask what was in the bottles.

1. Run **barium chloride + sodium sulfate** and copy all three equations.
2. Run **ammonium sulfate + barium nitrate** and copy all three equations.

Pair	Molecular	Net ionic
$BaCl_2 + Na_2SO_4$		
$(NH_4)_2SO_4 + Ba(NO_3)_2$		

a. The molecular equations are different. The net ionic equations are identical. Explain why, in terms of which species survive to the net ionic line.

b. Name a *third* pair of soluble compounds, not in the menu, that would give the same net ionic equation. Justify both of your choices with a solubility rule.

c. Here is a shortcut that looks consistent: in the complete ionic equation, split *every* formula into ions, including $BaSO_4$. Do that below, then cancel everything that appears on both sides. What is left?

Now compare with the line the demonstration writes. Which species does it refuse to split, and what physical fact about that species justifies the refusal?

> Split only what is genuinely dissolved and genuinely a strong electrolyte. A precipitate is a solid sitting on the bottom, so it keeps its whole formula. The same holds for gases, for liquid water, and for weak acids and weak bases such as $HC_2H_3O_2$ or NH_3, which stay mostly intact molecules in solution. Only strong electrolytes labelled (aq) get pulled apart.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it, check yourself, and write down where anything went off.

1. Aqueous potassium carbonate is mixed with aqueous calcium nitrate. Write the balanced molecular, complete ionic and net ionic equations, with state labels. If no reaction occurs, say so and justify it.

2. Aqueous sodium iodide is mixed with aqueous lead(II) nitrate. Write the net ionic equation and name the spectator ions.

3. Aqueous ammonium chloride is mixed with aqueous sodium nitrate. Predict the outcome, and explain — without doing any arithmetic — what the complete ionic equation would look like and why there is no net ionic equation to write.

4. $BaSO_4$ has K_{sp} = 1.1×10^{-10} at 25 °C. Calculate its molar solubility in pure water. Then explain whether calling $BaSO_4$ "insoluble" is honest, and state one assumption your calculation made.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet uses.

- Use **Random pair** ten times in a row and watch the **Predictions right** readout. Keep a note of which solubility rule you reach for wrongly — the exceptions row is usually the one doing the damage.
- Pour every pair that precipitates and rank the five solids twice: once by **K_{sp} at 25 °C** and once by **Molar solubility**. The two orders are not the same. Work out which solids swap places and what about their formulas causes it.
- Two of the solids come with an extra note about pH in the K_{sp} panel. Find them, and work out why adding acid pulls those particular anions out of the equilibrium when it does nothing to a sulfate or a halide.